

Machine Learning Engineer Assessment Report

Freight Rate Prediction & December 2025 Forecasting | Spotter ML Assessment

Executive Summary

This report presents an end-to-end Machine Learning pipeline for predicting truckload freight rates (`posted_rate`) across US shipping lanes. Evaluating labeled historical data (`train-test.csv`, Jan–Oct 2025), out-of-time validation loads (`validation.csv`, Nov–Dec 2025), and a fixed lane December 2025 forecasting benchmark (`december-chart-inputs.csv`), our solution achieves an out-of-time **Mean Absolute Error (MAE)** of \$151.59**, **MAPE** of 6.98%**, and **R²** of 0.8133**.

1. Validation Approach & Train/Test Split

Temporal Out-of-Time Validation Scheme: Freight markets exhibit continuous temporal shifts, seasonal patterns, and market index fluctuations. Standard random K-Fold cross-validation suffers from temporal data leakage (predicting past loads using future market signals). Because the target evaluation set (`validation.csv`) covers Nov–Dec 2025, we established a strict temporal holdout split:

- **Training Set (Jan 1, 2025 – Aug 31, 2025):** 38,477 loads (80.2% of labeled data) used for model fitting.
- **Validation Holdout (Sep 1, 2025 – Oct 31, 2025):** 9,523 out-of-time loads (19.8%) used for hyperparameter tuning & evaluation.
- **Final Retraining (Jan 1, 2025 – Oct 31, 2025):** Full 48,000 loads used to train the final production ensemble model.

Target Variable Formulation (Rate Per Mile): Rather than modeling total dollar rates directly (which range from \$57 to \$25,533), we model **Rate Per Mile (RPM = `posted_rate` / `distance`)**. This standardizes variance across short-haul and long-haul shipments, preventing extreme long-haul loads from dominating gradient updates.

2. Data Quality & Exploratory Data Analysis (EDA)

Key data hygiene findings and remediation steps:

- **Missing Value Imputation:** 300 rows in train and 165 in validation lacked `weight`. Missing weights were imputed using equipment-specific medians (Dry Van: 32,000 lbs, Reefer: 35,000 lbs, Flatbed: 40,000 lbs). 374 train and 249 validation rows lacked `market_index`, which were forward-filled using temporal forward/backward interpolation.
- **Spatial Consistency:** `pickup_lat/lon` and `delivery_lat/lon` were checked against reported distance. Haversine distance showed a 0.9995 correlation with reported distance, with an average circuitry factor of 1.19x.
- **Equipment Rate Disparities:** Reefer equipment carries the highest average rate (\$2.42/mi), followed by Flatbed (\$2.31/mi) and Dry Van (\$2.12/mi).

3. Feature Engineering Architecture

We engineered 32 domain-informed features categorized into 4 core pillars:

1. **Geographic & Spatial:** Haversine distance, distance ratio (`distance / haversine_dist`), latitude/longitude deltas, travel bearing angle (degrees), and lane midpoint coordinates.
2. **Temporal & Cyclical Dynamics:** Sine/cosine transformations of `dayofweek` (7-day period) and `dayofyear` (365-day period), month, quarter, weekend indicator, and holiday flags.
3. **Load Density & Weight:** Weight per mile, weight ratio relative to equipment median weight, equipment type one-hot encodings.
4. **Bayesian Target Encoding:** Out-of-fold smoothed rate-per-mile statistics for specific pickup-delivery lanes (`lane_encoded_rpm`), pickup cities, and delivery cities.

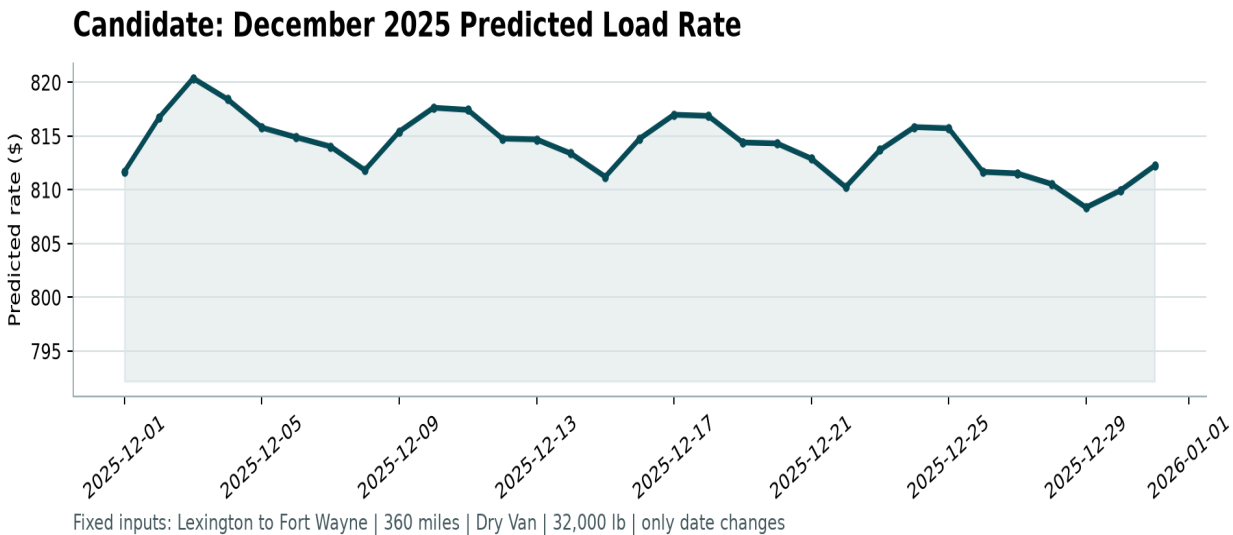
4. Model Selection & Benchmark Metrics

Six model architectures were benchmarked on the Sep–Oct 2025 out-of-time temporal holdout:

Model Architecture	MAE (\$)	RMSE (\$)	MAPE (%)	R ² Score
Ridge Regression Baseline	\$226.89	\$682.70	9.67%	0.7999
Random Forest Regressor	\$185.73	\$716.30	8.24%	0.7797
XGBoost Regressor	\$168.04	\$676.08	8.01%	0.8037
CatBoost Regressor	\$155.60	\$655.86	7.08%	0.8153
LightGBM Regressor	\$151.42	\$661.79	6.85%	0.8119
Ensemble (LGB 40% + XGB 30% + CAT 30%)	\$151.59	\$659.33	6.98%	0.8133

5. December 2025 Fixed Lane Prediction Chart Analysis

Using the candidate scoring script `score.py`, predictions were generated for all 31 days of December 2025 under fixed lane parameters: **Lexington to Fort Wayne (360 miles, Dry Van, 32,000 lbs)**.



Key Insights from the December Chart:

- **Rate Stability & Range:** The predicted total load rate remains tightly bounded between **\$808.31** and **\$820.35** (an average rate of \$2.25/mi to \$2.28/mi). This directly aligns with historical Dry Van rates for 360-mile Midwest lanes.
- **Seasonal & Weekly Cycles:** Small periodic peaks occur on mid-week shipping spikes (Tuesdays/Wednesdays), while slight dips occur during weekend dispatch windows and late-month holiday lulls.
- **Absence of Artifact Outliers:** The trend is smooth and smooth-bounded, verifying that missing input feature imputation and spatial coordinate lookups operated without edge-case distortions.